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PRACTICAL APPLICATION

Using Generative Artificial Intelligence in Health Economics and Outcomes Research: A Primer on Techniques and Breakthroughs

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Abstract

The emergence of generative artificial intelligence (GenAI) offers the potential to enhance health economics and outcomes research (HEOR) by streamlining traditionally time-consuming and labour-intensive tasks, such as literature reviews, data extraction, and economic modelling. To effectively navigate this evolving landscape, health economists need a foundational understanding of how GenAI can complement their work. This primer aims to introduce health economists to the essentials of using GenAI tools, particularly large language models (LLMs), in HEOR projects. For health economists new to GenAI technologies, chatbot interfaces like ChatGPT offer an accessible way to explore the potential of LLMs. For more complex projects, knowledge of application programming interfaces (APIs), which provide scalability and integration capabilities, and prompt engineering strategies, such as few-shot and chain-of-thought prompting, is necessary to ensure accurate and efficient data analysis, enhance model performance, and tailor outputs to specific HEOR needs. Retrieval-augmented generation (RAG) can further improve LLM performance by incorporating current external information. LLMs have significant potential in many common HEOR tasks, such as summarising medical literature, extracting structured data, drafting report sections, generating statistical code, answering specific questions, and reviewing materials to enhance quality. However, health economists must also be aware of ongoing limitations and challenges, such as the propensity of LLMs to produce inaccurate information ('hallucinate'), security concerns, issues with reproducibility, and the risk of bias. Implementing LLMs in HEOR requires robust security protocols to handle sensitive data in compliance with the European Union's General Data Protection Regulation (GDPR) and the United States' Health Insurance Portability and Accountability Act (HIPAA). Deployment options such as local hosting, secure API use, or cloud-hosted open-source models offer varying levels of control and cost, each with unique trade-offs in security, accessibility, and technical demands. Reproducibility and transparency also pose unique challenges. To ensure the credibility of LLM-generated content, explicit declarations of the model version, prompting techniques, and benchmarks against established standards are recommended. Given the 'black box' nature of LLMs, a clear reporting structure is essential to maintain transparency and validate outputs, enabling stakeholders to assess the reliability and accuracy of LLM-generated HEOR analyses. The ethical implications of using artificial intelligence (AI) in HEOR, including LLMs, are complex and multifaceted, requiring careful assessment of each use case to determine the necessary level of ethical scrutiny and transparency. Health economists must balance the potential benefits of AI adoption against the risks of maintaining current practices, while also considering issues such as accountability, bias, intellectual property, and the broader impact on the healthcare system. As LLMs and AI technologies advance, their potential role in HEOR will become increasingly evident. Key areas of promise include creating dynamic, continuously updated HEOR materials, providing patients with more accessible information, and enhancing analytics for faster access to medicines. To maximise these benefits, health economists must understand and address challenges such as data ownership and bias. The coming years will be critical for establishing best practices for GenAI in HEOR. This primer encourages health economists to adopt GenAI responsibly, balancing innovation with scientific rigor and ethical integrity to improve healthcare insights and decision-making.

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Key Points for Decision Makers

This primer introduces health economists to the essentials of using generative artificial intelligence (GenAI) tools, particularly large language models (LLMs), and their potential to streamline labour-intensive health economics and outcomes research (HEOR) tasks such as literature reviews, data extraction, and economic modelling. While chatbot interfaces provide an accessible entry point, more advanced use requires familiarity with application programming interfaces (APIs), prompt engineering techniques, and retrieval-augmented generation to enhance accuracy and relevance.

To effectively use LLMs in HEOR, health economists must understand their limitations – including risks such as hallucinations, security concerns, reproducibility challenges, and bias—and adopt best practices such as transparency, robust documentation, and secure deployment in compliance with regulatory standards.

As artificial intelligence adoption in HEOR grows, developing knowledge and expertise in LLMs will allow health economists to leverage their benefits responsibly while ensuring scientific rigor, ethical integrity, and compliance with regulatory standards.

1 Introduction

The emergence of artificial intelligence (AI) technologies has the potential to enhance key aspects of health economics and outcomes research (HEOR) by improving efficiency and streamlining certain tasks [1–6]. Generative artificial intelligence (GenAI), in particular, offers promising applications such as synthesising and analysing evidence, supporting clinical trial design, analysing trial data, and assisting with aspects of economic modelling [2, 3, 5, 6]. However, while these advancements highlight the technology's potential, their practical implementation in HEOR remains in the early stages, requiring further refinement and validation before their use in HEOR is commonplace.

To navigate this evolving landscape, it is essential for health economists to develop a foundational understanding of GenAI technologies and how these tools can complement their work. This primer is designed to introduce health economists, especially those with little or no experience with GenAI, to the basics of using these tools, particularly large language models (LLMs), in HEOR projects. After reading this primer, a reader should:

1. Have a basic understanding of AI and how GenAI can be useful for HEOR projects.
2. Be familiar with the basics of getting started with GenAI in HEOR, including how to interact effectively with LLMs through application programming interface (API) calls, prompt engineering, and understanding key principles of ethics, security, and reporting.
3. Understand the potential implications of adopting GenAI in HEOR and what this could mean for the future of the field.

The primer also provides practical recommendations to support researchers, using examples to demonstrate interactions with LLMs. These interactions are written in Python, which has been chosen due to its robust ecosystem for GenAI, including high-quality documentation, a wide selection of libraries, and an active developer community [7–11]. It should be noted that while this primer is focused on GenAI, it specifically refers to LLMs throughout since these foundation models are currently the most applicable branch of GenAI for health economics.

2 GenAI and LLMs

AI is a broad field focused on creating computational systems capable of performing tasks that typically require human intelligence, such as reasoning, learning, and problem-solving [12]. Within this domain, machine learning (ML) is a general approach where algorithms learn from data to make predictions or decisions, progressively improving their performance without being explicitly programmed for each specific task [12]. A significant subset of ML is deep learning (DL), which uses artificial neural networks with multiple layers—mirroring the complexity of the human brain—to model and understand complex patterns in large datasets (Fig. 1) [12].

Within DL, GenAI refers to a set of techniques centred on training and using foundation models [13]. These models are trained on extensive and diverse datasets, enabling them to generate new content across multiple domains such as text, images, or audio. Foundation models act as adaptable bases that can be fine-tuned for a wide range of specific tasks, and LLMs are a specific type of foundation model focussed exclusively on text-based data [14, 15]. By processing vast amounts of text, LLMs understand context and meaning, enabling them to generate human-like language. This

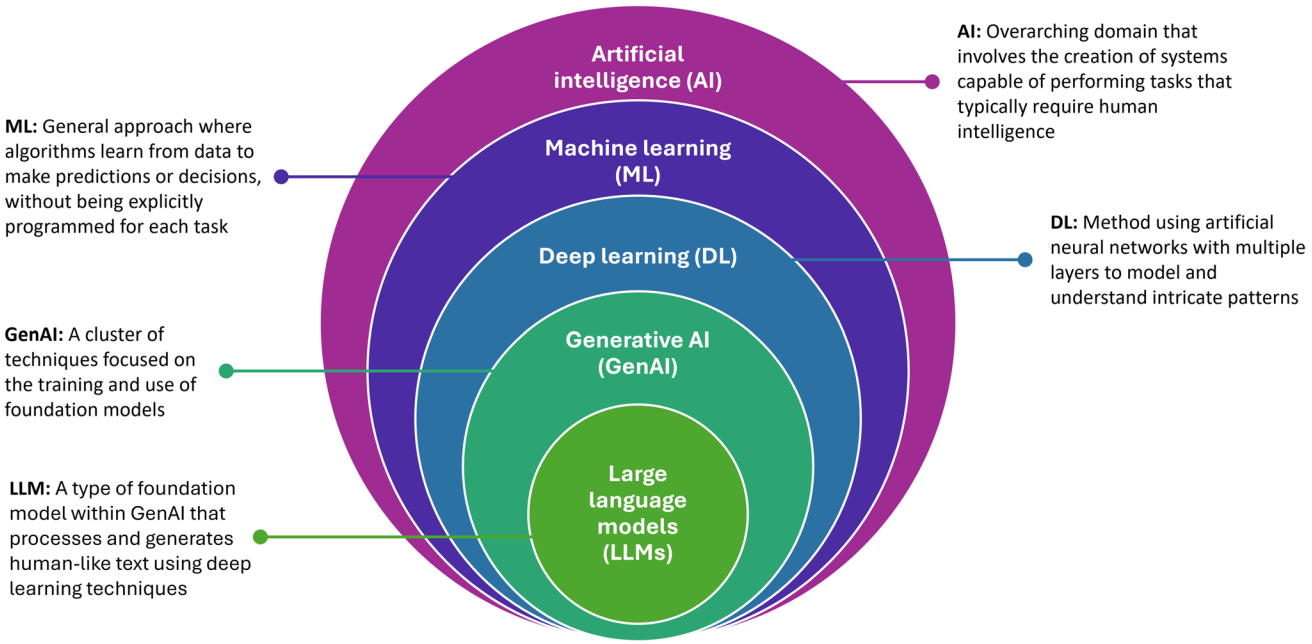


Fig. 1 Hierarchical relationship between artificial intelligence (AI), machine learning (ML), deep learning (DL), generative AI (GenAI), and large language models (LLMs), illustrating how LLMs are a specialised application of GenAI within DL. Adapted from Shahab et al., 2024 [79]

capability supports tasks such as translation, summarisation, and content creation.

GenAI and, in particular, LLMs are a recent advancement in AI and natural language processing that has captured public attention and sparked a wave of innovation. While previous iterations of ML and AI focussed primarily on classification and prediction tasks, GenAI emphasises the generation of new content. Some of the differences between 'traditional' AI and GenAI are shown in Table 1.

The growing popularity of GenAI can be attributed to several factors, such as reduced barriers to entry due to pre-trained models and its versatility across a wide range of applications—unlike traditional AI models, which are typically created for a single use case. GenAI is also easy to access through user-friendly interfaces such as ChatGPT [16].

As a subset of DL models, LLMs are distinguished by their pre-training on vast datasets and complex neural architectures. For example, in 2024, Meta revealed that Llama 3.1 was trained on 15 trillion tokens (i.e. units of text that the model processes individually), while Qwen 2.5 announced that its training dataset included up to 18 trillion tokens [17, 18]. This extensive pre-training allows LLMs to generate human-like language across a wide range of domains, making them more adaptable and flexible than traditional ML models, which, although reliable for specific tasks like coding or solving mathematical problems, tend to be trained for narrower, more specialised functions.

An additional unique feature of LLMs that distinguishes them from earlier AI approaches is their ability to scale through few-shot or zero-shot learning. Few-shot learning means models can perform tasks with minimal task-specific training, while zero-shot learning enables them to handle tasks with no prior task-specific training [19–21]. However, LLMs are also flexible enough to be used with highly complex prompt engineering strategies if the case requires, allowing for even greater adaptability in addressing specific tasks.

Several LLMs have emerged in recent years, each with its own strengths and applications. Among the most notable are GPT, Claude, Llama, and Gemini [22–25]. When discussing LLMs, distinguishing between open-source and closed-source models is crucial. Open-source LLMs have publicly accessible source code; users can freely use, modify, and distribute the model's architecture and weights freely, encouraging community collaboration and transparency [26]. Closed-source LLMs restrict access to these elements, maintaining them as proprietary and confidential [26]. These differences are important to consider in the context of HEOR and health technology assessment (HTA), where transparency, reproducibility, and confidentiality are paramount.

To date, the development of LLMs has been driven by closed-source models from companies like OpenAI, Anthropic, and Alphabet, who have the resources to invest in the training and hosting of LLMs. Previously, closed-source models such as GPT-4o, o1, and Claude 3.5 Sonnet

Table 1 Key differences between traditional AI and GenAI across key domains

Domain	Traditional AI	GenAI
Training process	Requires large amounts of data and is designed for specific use cases	Requires massive datasets and extensive training, focused on predicting the next word or token across diverse contexts
Learning process	Designed for specific tasks with algorithms optimised for targeted objectives	Uses a blend of supervised, unsupervised, and semi-supervised learning[a] to recognise patterns and predict outputs across diverse contexts
Dataset requirements	Often needs task-specific, labelled data (e.g. labelled images for image recognition)	Relies on massive, diverse datasets
Costs	High for each specific task	Initially expensive due to data requirements, but cheaper per use after initial set up
Computation resources [74–76]	Training and operation are relatively moderate in resource demand	Significantly more computationally expensive, especially during training, requiring extensive processing power and infrastructure, with potential environmental impacts (e.g. large-scale data centres)
Scope	Typically focussed on a single or few use cases (e.g. a model trained for conducting literature reviews of efficacy data cannot be used to review technology appraisals)	Broad applications (e.g. the same model can write poetry or code, like an R script for an NMA)
Purpose	Primarily used for tasks like classification and prediction	Suited for creating new content, including text, images, or code
Memory and context	Limited adaptability based on feedback; can incorporate user input or historical data for incremental improvements, but lacks dynamic, ongoing memory across interactions	Maintains context across extended interactions, allowing for coherent and complex outputs, potentially leading to more diverse application
Key risks/ethical concerns	Bias in training data; security; limited outputs/conclusions based on training data	Generating misleading or incorrect content, privacy and consent (especially with regards to medical data), ownership/attribution to creators, energy consumption
Responsible parties	Often developed and maintained by organisations or research institutions	Primarily created and maintained by large tech companies and open-source communities
	Typically, proprietary and paid for through licensing or project-based costs	Access via APIs, subscriptions, or enterprise licences

AI artificial intelligence, *API* application programming interface, *GenAI* generative AI, *NMA* network meta-analysis

[a]Supervised learning: models are trained using labelled data, where each input has a corresponding output. Unsupervised learning: models are trained on unlabelled data to identify patterns or groupings. Semi-supervised learning combines labelled and unlabelled data; leverages unlabelled data to improve supervised learning tasks when labelled data are scarce or expensive [77, 78]

were among the best performing LLMs based on widely established benchmarks [27]. However, there has been significant progress in the development of open-source models, with Deepseek in particular now outperforming its closed-source counterparts, according to established leaderboards and metrics, such as quality (i.e. accuracy and completeness of a model's responses), latency (i.e. time taken for a model to process a prompt and start to generate a response), and output speed (i.e. the rate at which a model can generate an entire response once it starts generating) (testing as of January 2025) [28, 29]. Furthermore, it has been reported that Deepseek was trained at a fraction of the cost of the closed-source models, though this is the subject of considerable debate [30–32]. Beyond performance, the availability of model weights and training data associated with some open-source models may present unique opportunities in the context of HEOR and HTA [33].

While text-based LLMs have dominated the AI landscape, recent advancements have led to the development of multimodal models such as GPT-4, which are capable of processing and generating content across different formats, such as text, images, and audio [22, 34, 35]. These models are an evolution of text-based LLMs and built on the same architecture, but trained on diverse datasets, making them more flexible and versatile. This opens up exciting possibilities for HEOR, particularly in handling complex data [35, 36]. For example, in healthcare settings, multimodal models can integrate medical imaging with electronic health records to enhance disease diagnosis, treatment planning, and patient monitoring [37]. In HEOR,

similar capabilities could enhance the analysis of real-world evidence by identifying treatment adherence patterns or long-term patient outcomes in diverse data sets, for instance. Although a detailed discussion is beyond the scope of this paper, the potential applications of multimodal GenAI are worth acknowledging.

3 Applications of LLMs in HEOR

For health economists new to GenAI technologies, the most accessible way to explore LLMs is through chatbot interfaces, many of which are publicly available and free to use (e.g. ChatGPT [chatgpt.com], Claude [claude.ai], Meta AI [meta.ai], Google Gemini [gemini.google.com], Microsoft Copilot [copilot.microsoft.com]) [8, 22, 24, 25, 38]. These chatbots offer a straightforward introduction to LLM capabilities, enabling health economists to experiment with prompt engineering and gain a basic understanding of how to interact with LLMs for specific tasks. For health economists, this could involve simple tasks such as asking the chatbot to summarise research papers, screen a limited number of abstracts, or interpret the results of an economic model. These activities can help users become confident in interacting with LLMs and more fully understand their potential use in everyday HEOR tasks.

While chatbots provide a convenient and easily accessible starting point, their use in conducting complex or large-scale HEOR projects is limited. The design of the chatbot interfaces is inherently limited; they cannot easily scale or link multiple LLM processes together. Chatbots also restrict the use of advanced prompt engineering techniques, such as self-reflection or self-consistency, which are necessary for generating accurate HEOR results. Additionally, they do not allow access to the LLM's settings (hyperparameters), which are needed for optimising performance.

For example, screening hundreds or even thousands of abstracts using a chatbot interface would require manually copying and pasting each abstract into the chatbot and recording the results—a process unlikely to confer significant time and cost savings compared with traditional manual methods [1]. Similarly, while using chatbots to write code for building health economics models is technically possible, it would be laborious and difficult to test and replicate into other settings [3, 4]. Privacy concerns are also a significant consideration when processing sensitive health data through public chatbot interfaces; privacy is discussed further in Section 5.

To overcome these limitations and enable health economists to fully leverage LLMs in HEOR workflows, APIs offer a more robust solution via direct programmatic access to LLM functionalities, enabling seamless integration into custom applications and workflows. An API is an interface that allows different software applications to communicate and interact with each other [39, 40]. It defines the methods and data formats that programs can use to request and exchange information, enabling the integration and functionality of various software components or services. In essence, an API is a means by which a software developer—in this case, the developers of LLMs—can offer access to the core functionality of that software through specific 'endpoints' without providing access to source code. Another benefit of APIs is that they can enable secure and efficient access to LLM capabilities while keeping proprietary datasets or sensitive information private [41].

This approach offers several advantages for health economics applications, including scalability to process large volumes of data efficiently, the ability to programmatically link distinct LLM interactions together using business logic, and the capability to integrate LLMs into existing software. Linking LLM interactions through business logic facilitates a toolchain approach that allows for much more diverse and complex uses of LLMs for HEOR purposes. Figure 2 illustrates an API call using GPT-4o and Python, demonstrating a basic method of programmatically interacting with an LLM through simple input/output (I/O) prompting. More refined prompting strategies and methods for working with LLMs are described in subsequent sections.

By starting with chatbot interfaces to build familiarity and progressing to API-based solutions for advanced applications, health economists can develop a foundational understanding of LLMs and their potential to enhance HEOR research.

3.1 How to Work Best with LLMs

To make the most of LLMs and effectively integrate these tools into HEOR workflows, it is essential for health economists to understand what LLMs are—and perhaps more importantly, what they are not. LLMs are highly advanced 'next token predictors', not traditional lookup tools or databases. Tokens are basic units of text that the model processes (see Table 2 for common GenAI terms and definitions) [42]. As a general guideline, one token is approximately 4 characters or 0.75 words for English text [42]. LLMs learn to recognise and understand the statistical relationships between sequences of tokens, allowing them to generate text. They do this by identifying patterns learned from vast amounts of training data and probabilistically predicting the next token in a sequence based on the context provided by the user [43, 44]. Understanding this distinction is essential for health economists: LLMs do not access a fixed repository of information outside of the data they were trained on. While some chatbot interfaces, such as Bing, combine LLMs with

```
import openai
response = openai.ChatCompletion.create(
  model='gpt-4o',
  messages=[
        {"role": "system", "content": "You are a helpful health economist" },
        {"role": "user", "content": "Describe the role of a health economist \
         in 10 words"},

    ],
  temperature=0
)

print(response["choices"][0]["message"]["content"])

#Response: Analyzes healthcare systems to inform policy, allocate resources,
#          improve care.
```

Fig. 2 Example of an application programming interface (API) call using Generative Pre-trained Transformer 4o (GPT-4o) and Python

Table 2 Common key terms and definitions used in generative artificial intelligence (GenAI)

Term	Definition
Large language model (LLM)	A type of foundation model within GenAI trained on massive text-based datasets, enabling them to generate human-like language
Token	A basic unit of text that is processed by LLMs. One token is approximately 4 characters or 0.75 words of English text
Prompt	The text provided to an LLM to elicit a response. Prompts significantly affect the quality of outputs from an LLM
Zero-shot prompting	A simple type of prompt where the model is asked to respond without any prior examples and instead to rely purely on its training dataset
Few-shot prompting	A prompting technique in which the model is given a few examples in the prompt before being asked to perform a similar task, which can lead to more accurate responses
Application programming interface (API)	An interface that allows different software applications to communicate with another by defining methods and data formats that programs can use. LLMs can be accessed via APIs
Hallucination (in GenAI)	A response from an LLM that is inaccurate or fabricated information, most likely due to a lack of sufficient or accurate information in its training data
Retrieval-augmented generation (RAG)	A technique that allows a foundation model such as an LLM to retrieve factual information from external sources rather than relying solely on their training dataset to generate a response

live internet searches to provide real-time information, most LLMs operate solely on their training data [45].

As such, LLMs generally work better in very narrow, specific contexts where all of the information to answer the question is provided and the question asked is very specific. Asking an LLM overly broad questions or to infer information that is not explicit can lead to suboptimal results. 'Task decomposition', or breaking down complex questions into smaller, manageable parts, is recommended to improve the accuracy and reliability of the LLM's responses [46, 47].

3.2 LLMs' Strengths in HEOR Applications

LLMs have considerable potential for use in HEOR tasks that leverage their pattern recognition and text-generation capabilities. Research has shown that LLMs can be applied to the following common HEOR applications:

1. *Summarising:* LLMs can condense and organise key findings from clinical trial publications, clinical study reports, HTA reports, and other medical literature [2, 3, 48]. This also includes producing plain language summaries of complex medical literature.

2. *Extracting:* LLMs can pull structured data, such as PICO (population, intervention, comparator, and outcome) elements from clinical trial documents [3, 5, 48].
3. *Writing:* LLMs can draft sections of HEOR reports, such as the results of health economics analyses, including network meta-analyses (NMAs) [3].
4. *Coding:* LLMs can generate code for statistical analyses and economic modelling (e.g. R scripts for NMAs) [2–4, 49].
5. *Answering one question:* LLMs perform best when completing one 'job' at a time. 'One API call for one task' is a good rule of thumb when working with LLMs through APIs [50, 51].
6. *Reviewing:* LLMs can act as an additional reviewer to enhance the quality of materials generated by humans or other AI tools [2, 52, 53].

Starting with these well-defined tasks allows health economists to experiment with LLMs, explore their capabilities, and determine their usefulness for specific needs. In general, health economists may find it beneficial to focus on the use cases and heuristics outlined above when interacting with LLMs, whether through chatbots or APIs (as we recommend) in order to get the best out of the models. Additionally, following best practices with regards to prompt engineering and providing the LLM with the necessary context (see Section 4 for further details) can improve the quality of responses [54]. A 'zero-shot' approach, for instance, is unlikely to achieve the desired results, particularly in a field as complex as HEOR.

3.3 Recognising LLM Limitations

Effectively using LLMs requires health economists to recognise and understand their limitations. The quality of an LLM's response depends heavily on the model's training data. Where the model lacks sufficient information, 'hallucinations'—inaccurate or misleading information—may be generated. This issue is compounded by the fact that LLMs are trained on vast datasets from the internet, which can inherently contain incorrect, biased, or unverified information. This can potentially lead to LLMs unintentionally generating and propagating falsehoods on a large scale. Techniques such as retrieval-augmented generation (RAG) can help mitigate this by enabling the model to retrieve factual information from external sources, such as policy documents or clinical guidelines, before generating a response [55]. For example, Liu et al. have demonstrated that incorporating RAG significantly improved accuracy in a systematic literature review and meta-analysis (1.35 odds ratio increase compared with using LLMs alone) [56].

While LLMs can provide insights based on statistical patterns in their training data, they still lack true understanding of complex concepts. Their ability to perform strategic planning and or solve problems autonomously is an active field of research and development. For example, initiatives like the OpenAI o1 series are incorporating multi-step reasoning directly into its design to improve these functions [57]. Although rapid innovation may lead to breakthroughs, LLMs are currently best seen as tools that can complement rather than replace human expertise and labour.

4 Prompt Engineering

To fully leverage LLMs in HEOR, mastering the skill of writing effective prompts—a growing field known as 'prompt engineering'—is crucial [58]. A prompt is the text provided to an LLM to elicit a specific response and is separate from other parameters and hyperparameters like model name or temperature. Prompts can range from simple questions, known as basic zero-shot I/O prompting, to more advanced methods such as 'few-shot' and 'chain-of-thought' prompting, which provide context and aim to simulate complex reasoning [59–62].

With few-shot prompting, the LLM is provided with a small number of examples before being asked to perform a similar task, which can lead to more accurate responses. Chain-of-thought prompting encourages the LLM to break down its reasoning process, delivering intermediate steps in its analysis or decision-making. Recent developments in LLMs, such as OpenAI's o1 series model, have introduced significant advancements in the use of chain-of-thought prompting [57]. For health economists, understanding when and how to apply these different techniques to a given situation is essential [58].

Health economists can use basic zero-shot I/O prompting to ask the LLM to perform a straightforward task, such as extracting key information from abstracts. For example, a simple API call—'Extract the population of interest from this abstract'—instructs the LLM to identify relevant study populations without requiring specific examples or additional context. While this approach may appear overly simplistic, it is perfectly suited to this task and demonstrates an important principle: as with economic models, health economists should choose the most parsimonious approach when working with LLMs to minimise unnecessary complexity and improve reliability in the results.

However, there are many situations where simple zero-shot I/O prompting is likely to fall short due to the field's complexity and the need for domain-specific knowledge. For more complex HEOR scenarios, a simple prompt like 'Build an economic model in non-small cell lung cancer' is unlikely to produce a useful or accurate response. In this example, prompts that provide examples, context, code, or an underlying process that can logically guide the LLM are needed [3, 4]. In these cases, few-shot prompting, potentially in combination with RAG, can be particularly effective. RAG is well-suited for tasks requiring data or knowledge that fall outside the model's training set—whether due to the training data cut-off date, proprietary restrictions, or the need for highly specialised information that may not be included in public data [63]. When used alongside other prompt engineering techniques, RAG can enhance the LLM's ability to retrieve up-to-date or domain-specific information as part of its response generation [63]. Figure 3 demonstrates how a few-shot prompt designed for calculating drug acquisition costs provides better structure and clarity compared with a basic I/O zero-shot prompt.

For more complex tasks, such as budget impact calculations, chain-of-thought prompting can generate more accurate results by helping the LLM break down its reasoning for each step. Figure 4 illustrates a chain-of-thought prompt that guides the LLM through each phase of a budget impact calculation.

By employing advanced prompting techniques such as few-shot and chain-of-thought prompting and RAG, health economists can more effectively harness the capabilities of LLMs. Although other more advanced prompting strategies are available, a detailed exploration of these techniques is beyond the scope of this primer.

```
import pandas as pd
import openai

data = pd.read_csv(path)

prompt = f"""Your question is: 'Write an R script that calculates week-on-week drug acquisition costs over a 2
year period for medication X, based on the provided data (demarked by [Data])'

[Data]
{{data}}

Before you answer your question, consider the following example, demarked by &&&&&.

&&&&&

Example question: 'Write an R script that calculates week-on-week drug acquisition costs over a 2 year period
for medication Y. A 100mg vial of medication Y is priced at $300. The dosing schedule is 200mg per 2 weeks.'

Example answer:

y_price_per_vial <- 300
dose_per_2_weeks_mg <- 200
mg_per_vial <- 100
total_weeks <- 52*2

cost_per_2_weeks <- (dose_per_2_weeks_mg / mg_per_vial) * y_price_per_vial

weekly_costs <- numeric(total_weeks)

for (i in 1:total_weeks){{
    if (i%%2 == 0) {{
        weekly_costs[i] <- cost_per_2_weeks
    }} else {{
        weekly_costs[i] <- 0
    }}
}}

&&&&&

Now, answer your question. Let's think in steps to ensure we get the right answer."""

response = openai.ChatCompletion.create(
    model='gpt-4',
    messages=[
        {"role": "system", "content": "You are an expert health economist"},
        {"role": "user", "content": prompt}
    ],
    temperature=temperature,
)
```

Fig. 3 Example of a few-shot prompt for calculating drug acquisition costs

```
import pandas as pd
import openai

data = pd.read_csv(path)

prompt = f"""Your question is: 'Analyse the budget impact of introducing a new hepatitis C treatment, based on
the provided data (demarked by [Data])'

To answer this question, take the following steps:
1. Estimate the target population size over a 5-year period (starting 2024).
2. Estimate market shares for the current scenario, and the future scenario.
3. Estimate annual per patient costs for each of the current treatments.
4. Estimate annual per patient costs for the new treatment.
5. Bring together results from prior steps to calculate and compare the total costs of the current and new
scenario, over the 5-year period.

[Data]
{data}

Now, answer your question. Let's think in steps to ensure we get to the correct answer.
"""

response = openai.ChatCompletion.create(
    model='gpt-4',
    messages=[
        {"role": "system", "content": "You are an expert health economist"},
        {"role": "user", "content": prompt}
    ],
    temperature=temperature,
)

output = response.choices[0].message.content
```

Fig. 4 Example chain-of-thought prompt for budget impact calculation

5 Security and Risk

Implementing new technologies in the pharmaceutical industry requires robust security. In HEOR, and particularly for HTAs, data are likely to be confidential or embargoed and may even include anonymised patient-level data. While it is possible to deploy and interact with LLMs securely, health economists must consider several factors such as data sensitivity (e.g. preserving confidentiality), legal compliance (e.g. the European Union's General Data Protection Regulation [GDPR] and the United States' Health Insurance Portability and Accountability Act [HIPAA]), the type of model to use (i.e. open vs closed source), and the available cybersecurity and development and operations (DevOps) expertise and infrastructure.

With this in mind, there are several options for securely deploying and interacting with LLMs in pharmaceutical or healthcare settings. This paper does not endorse a specific model type (open vs closed source) or deployment method but aims to educate potential LLM users regarding the advantages and limitations of each approach. It should be noted, as previously stated, that at the time of writing, open-source models are as good or in some cases outperforming closed-source models according to established performance metrics, though this is an ever changing picture [29].

One approach is to use local deployment, where the LLM and all associated data remain within the organisation's controlled environment. This allows the organisation to choose the LLM, manage their own servers, and interact with the LLM through their own firewall. This setup ensures that sensitive information never leaves the company's secure infrastructure and is a robust solution, assuming that the infrastructure itself is secure. However, this approach has some limitations. A key disadvantage is that it limits the LLMs that the analyst is able to access, since GPT-4o, Claude Sonnet, and Gemini are closed source and therefore not suitable for local deployment. Furthermore, it requires in-house staff with significant DevOps and cybersecurity expertise to provision and maintain the servers, as they must continually meet industry-standard cybersecurity measures such as Open Web Application Security Project (OWASP) or risk significant reputational and financial damage [64]. Moreover, interacting with LLMs via standard servers is likely to be prohibitively slow, meaning that deploying LLMs within an organisation's firewall will require access to high-end graphics processing unit (GPU) servers. Due to the high demand, the availability of such high-performance GPUs is currently limited. Depending on the size of the LLMs (often dictated by number of parameters), multiple high-performance servers may be needed to self-host them effectively. However, it is not known at this stage whether the trend will be towards smarter models with larger numbers of parameters or smaller models that can run on everyday devices such as smart phones and personal computers.

Another option for secure LLM deployment is to use APIs hosted by trusted providers who offer enterprise-grade security measures and compliance with data protection regulations such as International Organization for Standardization (ISO), HIPAA, and System and Organization Controls 2 (SOC 2). This approach has several advantages, as it provides access to the latest LLMs with the most robust security measures available, without the need to physically source GPU servers. Furthermore, working with an established provider such as Amazon Web Services (AWS), Google Cloud, or Microsoft Azure offers contractual protections regarding how data are stored and used. For example, it can be contractually agreed that data sent to the LLM will not be used to train future model iterations, a significant concern for businesses using proprietary or commercially sensitive data. However, there are also limitations to this approach. Although contractual obligations are reassuring, they are not foolproof, and realistically, the risks of interacting with LLMs through providers cannot be entirely eliminated. Furthermore, such arrangements could tie businesses into using specific closed-source LLMs. Additionally, the cost of interacting with closed-source LLMs can be significant, especially for high-volume tasks such as abstract screening for literature reviews. Since such costs can be prohibitive, it is recommended that analysts carefully forecast their likely usage before opting for hosted APIs.

A third, intermediate solution is for the analyst to download and host open-source APIs on cloud servers. This approach combines the benefits of open-source models with the contractual enforced security and data usage protections of hosted solutions, without the need to physically source expensive GPU servers. The disadvantages of this approach include increased requirements for cybersecurity and DevOps expertise (as the APIs are not hosted as a 'service'), lack of access to closed-source LLMs, and the cost of requisitioning high-end cloud-based GPU servers, which may be less cost-effective than using hosted APIs. While this approach is more accessible than fully self-hosting LLMs, the need for DevOps and cybersecurity expertise still creates a significant barrier to entry when hosting open-source models in this manner.

If cybersecurity expertise or access to the right equipment (either physically or through cloud-based solutions) is not available to the analyst, it is highly recommended to avoid working with confidential data. Without the necessary security measures, analysts may be forced to interact with LLMs through publicly available chatbots and APIs, where they have no physical or contractual control over the security protocols. This increases the risk of data breaches or unauthorised access to proprietary information, which could have severe financial and reputational consequences for the business. This has significant implications for analysis of sensitive data, such as patient-level data used in real-world evidence analyses or clinical trials, or confidential data on pharmaceutical pricing. In these cases, relying on publicly available chatbots and APIs poses a substantial risk of data privacy breaches and/or potential violations of legislative privacy acts such as GDPR and HIPAA. Analysts must carefully consider these risks when deciding the best way to access LLMs, if intended for handling confidential data.

The multiple risks and potential mitigations to consider when using LLMs for HEOR are summarised in Table 3.

6 Reproducibility, Transparency, and Reporting

Reproducibility and transparency are cornerstones of HTA and are fundamental for ensuring the credibility and reliability of research findings, which in turn support informed resource allocation decisions. When HTA reviewers are unable to understand or sufficiently reproduce the results of an analysis for HTA, the validity of the outputs may be significantly compromised. As health economists explore the use of LLMs in HEOR, it is essential to consider how these principles apply to GenAI-generated content.

The use of LLMs in HEOR presents unique challenges in this regard. LLMs can be used to generate diverse content, ranging from summarising the results of a cost-effectiveness analysis to generating executable code for running a model. This variability makes it difficult to determine when and how the role of an LLM in HEOR should be explicitly declared. For example, if LLM architecture and weights are directly used to run an NMA or cost-effectiveness analysis—a practice that is strongly discouraged—serious validity issues arise. The vast multi-billion parameters of LLMs make it impossible to fully understand its reasoning or how it arrived at specific answers. However, when LLMs are used to generate executable code for an economic model or NMA, the resulting script can and should be rigorously validated in the same way as if written by a human, since the content's purpose and format are indistinguishable, regardless of origin. Additionally, the involvement of an LLM in generating such content should be explicitly declared in any reporting.

Given the 'black box' nature of LLMs, ensuring transparency is particularly challenging. Clear declarations are essential to inform all stakeholders of the LLM's role in the process. Establishing a consistent approach for reporting the use of LLMs will help health economists ensure that AI-generated content meets reproducibility and validation standards in HEOR. Health economists using LLMs to conduct HEOR

Table 3 Summary of potential risks and mitigations when deploying LLMs for HEOR

Risk	Potential consequence	Risk classification	Mitigation
Confidential data sent to public server	Data leak leading to commercially sensitive information being in the public domain	High	Self-hosting LLMs on AWS (for example) Hosting LLMs on managed servers through contractual agreements (AWS, Google Cloud, Azure, etc.) Robust cyber security protocols and expertise [64] Providing users of LLMs with mandatory robust cyber security training and adherence to appropriate data protection and cybersecurity policies
Confidential data being used to train models	Confidential data emerging in output from LLMs due to being in training data	High	Managed contractual agreements with hosting providers that explicitly prohibit the use of input prompts to train models Self-hosting
Prompt injection	Text inserted into prompts to 'trick' AI into executing malicious code on servers	High	'Sandboxed' environments for LLMs, robust cyber security protocols, and expertise
Hallucinations	Inaccuracies in results propagating to HTA submissions, publications, etc. Using proprietary closed-source LLMs that are 'trained' to give biased results Users unaware of how results are generated, leading to reduced trust	Moderate	Robust prompting strategy, human-in-the-loop approach, RAG
Non-reproducible results			Robust and transparent reporting
Inaccurate results			Advanced prompt engineering, e.g. self-consistency approach with human-in-the-loop oversight
Biased results			Human-in-the-loop approach, robust validation protocols Asking the LLM to report exactly how results were generated (i.e. if applicable the 'chain of thought'/ reasoning steps)
Lack of transparency			Possible additional checks by other LLMs, or assigning the role of 'checker' to an agent-based LLM Transparently reporting exactly which LLM and version was used so that results can be replicated to some extent
Overreliance on LLMs	Diminished critical thinking by human reviewers Lack of opportunities for junior health economists to gain experience	Moderate	LLMs used to augment rather than replace HEOR specific tasks; human involvement must always be there Proper training given on how to use LLMs as a 'force multiplier' for HEOR-related tasks rather than a replacement

AI artificial intelligence, *AWS* Amazon Web Services, *HEOR* health economics and outcomes research, *HTA* health technology assessment, *LLM* large language model, *RAG* retrieval-augmented generation

projects should ensure validation and repeatability via the following steps:

1. *The exact version of the LLM used, including the date:* Clearly reporting the version makes it easier for reviewers to replicate results.

2. *A brief description of the prompting techniques used:* While it would be unreasonable to document every prompt used to generate content, providing a description of the prompting techniques and the contextual information provided should be standard. This allows reviewers to determine whether the most appropriate techniques were used and to potentially replicate the results as closely as possible.
3. *A comparison of the LLM's process relative to an established 'gold standard' or human benchmark:* When an LLM is used to generate content for HEOR, the accuracy of the process relative to a human benchmark performed on a subset of the problem or a similar problem should be presented alongside the content and results in the first use. Without this validation, it is impossible to evaluate whether the LLM's output is accurate, reliable, and robust.

In summary, while LLMs present new challenges in terms of transparency, validation, and reporting, these issues can be effectively addressed through a clear understanding of how LLMs work and by adhering to a pre-specified and well-defined reporting structure.

7 Validation and Repeatability

Validation and repeatability of LLMs is a nuanced topic with few definitive or straightforward answers. The integration of LLMs into HTA introduces challenges to repeatability and validation, some of which already exist, while others are unique to the application of LLMs in this context. LLMs are by design probabilistic and therefore incorporate randomness in their outputs, which can lead to variability in results even when the same input is used. While this variability may be subtle, it could have significant implications in HTA, where small differences in outcomes can influence policy and funding decisions.

This situation is somewhat analogous to managing uncertainty in statistical analyses and economic modelling for HTA. In both cases, randomness is used to quantify and understand modelling uncertainty either through the concept of repeated sampling in a frequentist analysis or via Monte Carlo simulations in Bayesian frameworks. Similarly, for LLMs, we can account for randomness by recognising the stochastic nature of their outputs and taking steps to ensure that the LLM converges on more predictable and reproducible responses.

Several strategies can be employed to address this challenge:

1. *Temperature setting:* The 'temperature' parameter in LLMs controls the randomness of the model's outputs. Lower temperature settings (closer to 0) result in more deterministic and consistent responses, while higher settings increase creativity and variability. For HTA applications requiring high reproducibility, a low temperature setting can help ensure more consistent outputs across multiple runs.
2. *Self-consistency approach:* Instead of relying on a single output from an LLM, a self-consistency approach generates multiple outputs for the same input and selects the most common or representative response [65]. This method can mitigate the impact of occasional outlier responses and provide a more stable output.
3. *Seed setting:* Some LLMs allow the use of a random seed. By using the same seed across different runs or implementations, researchers can ensure that the 'random' elements of the model behave consistently, thereby enhancing reproducibility.
4. *Using different LLMs:* Employing different LLMs for the same task and comparing outputs can provide insights into the consistency and reliability of the results. Convergence of findings across different models can increase confidence in the reproducibility of the outcomes.
5. *Choosing the right LLM:* Selecting LLMs according to their strengths, combined with effective prompting that is tailored to the task at hand, and the application of structured processes and business logic, can improve reproducibility.

By effectively applying these techniques, reproducibility can be substantially increased, and the occurrence of hallucinations can be minimised. It is highly recommended that these best practices are followed when interacting with LLMs, particularly within an HTA context.

8 Ethics

The ethical implications of using AI in HEOR, including LLMs, remain open and complex [2]. As health economists begin to explore these technologies, understanding when ethical concerns arise and how to address them is essential. For many routine tasks in HEOR, such as embedding a data array into an R script or changing an Excel value, ethical considerations are minimal to non-existent. However, ethics becomes a pertinent concern when AI is used more broadly—for example, to influence patient outcomes, regulatory decisions, or cost-effective analyses [66]. Health economists should carefully assess each use case

to determine the level of ethical scrutiny and transparency required.

For scenarios where ethics are a core consideration, determining accountability remains complex. For instance, if an LLM recommends the wrong treatment, who bears responsibility—the LLM developers, the researchers using the model, or the institutions making recommendations based on AI-driven solutions [66]? This question remains open, and there are no straightforward answers. Traditionally, accountability in HTA has been clearly delineated: manufacturers are responsible for the materials they submit, while HTA agencies are accountable for their evaluations and final decisions. However, it is worth noting that even with human experts, who are considered the gold standard for producing accurate and robust HEOR materials, mistakes are common [67]. Health economists using LLMs should consider how shared accountability can be managed to uphold the integrity of HTAs while mitigating the risks associated with using LLMs.

When adopting new technologies like LLMs in HEOR, the phenomenon of status quo bias and risk miscalibration—where people tend to overestimate the risks associated with new technologies while underestimating the risks of maintaining current practices—is especially relevant [68]. This bias can lead to an overly cautious approach to AI adoption, potentially delaying the realisation of its benefits in healthcare, such as expediting patient access to new medicines. The analogy of self-driving cars is apt; decision-makers may focus on a single death involving a self-driving car, giving it more weight than the approximately 1.2 million deaths caused by human drivers every year [69]. In HEOR, this extends to situations where errors in AI-generated deliverables, such as inaccuracies in data analysis or model inputs, are likely to be interpreted far more negatively compared with mistakes made by humans. Health economists should weigh concerns about inaccuracies in LLMs or GenAI against the risks of foregoing technologies that have the potential to enhance analyses and improve efficiency.

Therefore, while significant ethical questions on whether to adopt AI remain, we must also consider the ethical implications of *not* embracing AI for HEOR. In a field where timely and accurate analyses can directly impact patient access to life-saving treatments, delaying AI adoption could result in losses in quality-adjusted life years while patients wait for HEOR-related materials to be generated for HTA processes. To echo the recent position statement from the National Institute for Health and Care Excellence (NICE), however, given the importance of the decisions being made and their impact on patients, a human-in-the-loop approach should always be used [70]. Even where AI is used to produce materials, the final decision-making should always remain with humans.

Another key ethical consideration is potential bias within LLMs [66]. A common concern is that these models may have been trained to produce answers specifically geared towards specific outcomes, which could yield more or less favourable conclusions on questions of clinical and cost-effectiveness depending on the direction of bias. While this is a valid concern, at present not enough is known about whether such biases exist and, if they do, in which direction the bias goes. This means health economists must remain vigilant, recognising such biases if they arise on a regular basis and in predictable patterns during their work. Built-in guardrails in common closed-source models, such as reinforcement learning from human feedback, can help prevent certain types of bias and misinformation. Analysts can further mitigate potential issues by using pre-emptive testing and benchmarking. Additionally, RAG and proper prompt engineering can reduce hallucinations and bias.

A possible solution to this issue could be to use fine-tuned models that have been recalibrated to give more balanced answers to specific questions. However, this approach carries the risk of introducing new biases, potentially in the opposite direction or even creating bias that was not previously there, depending on the perspective of the analyst. Vigilance on this topic is strongly encouraged as the uptake of LLMs in HEOR grows.

Another related issue is the concept of LLMs being trained on data for which the developer does not own the intellectual property. This is a live issue subject to ongoing debate and litigation [71, 72]. While it may seem removed from the daily work of an analyst, it could have substantial implications for the future of AI in HEOR, especially if specific LLMs are found to have been illegally trained and subsequently discontinued.

Finally, another issue to consider is whether accelerating the development of HEOR materials for HTA would have knock-on effects in other areas of the system. The specifics of this discussion are beyond the remit of this primer, but this is an important consideration for the uptake of AI in HTA.

9 Future Considerations

As LLMs and other AI technologies continue to evolve, their potential to transform HEOR becomes increasingly apparent. For health economists new to these tools, understanding how GenAI could shape future developments presents an opportunity to explore potential applications in HEOR. Three key areas where GenAI shows promise are in developing and maintaining 'living' HEOR materials, patient empowerment, and enhanced analytics that may support more timely access to medicines.

9.1 Living HEOR Materials

The concept of 'living' HEOR materials represents a significant shift from traditional static reports and analyses. For health economists, this could mean leveraging GenAI, particularly LLMs, to create HEOR documents and models that could become dynamic, continuously updated resources that reflect the latest data and insights in real-time. This approach could change how health economic evidence is generated, disseminated, and utilised. For example, GenAI could continuously scan and integrate new clinical trial results, real-world evidence, and economic data into existing cost-effectiveness models. This would allow for the automatic updating of key metrics, such as incremental cost-effectiveness ratios, as new information becomes available. Such dynamic models and documents would continuously evolve by incorporating new data on effectiveness, safety, and costs, ensuring decision-makers have access to the most up-to-date evidence. This adaptability would support more timely and informed healthcare resource allocation decisions. This approach would be particularly valuable for assessing novel therapies, those granted accelerated approval, or for managed access agreements, where long-term data may not be initially available or where approvals are conditional on further evidence generation.

However, health economists should also consider potential challenges and limitations in developing living HEOR materials. One concern is the potential for a 'single source of truth' to emerge for materials produced for HTA submissions. While this could reduce bias by standardising information, it may fail to fully capture the heterogeneity present in different disease areas and outcomes for patients. Additionally, for living materials to function effectively, a single entity may need ownership of the data, which raises issues of control and influence, unless a model of 'distributed ownership' can be agreed. However, a distributed model also presents challenges, since one party might be perceived as having undue control over how the materials are produced and updated.

A transition to living HEOR materials and the creation of a 'single source of truth' would also require additional time investment from health economists to validate the ground truth materials, ensuring they can be used by others with confidence. A similar point arises when considering the implementation of GenAI into HEOR workflows generally. The upfront time investment required varies depending on the stakeholder. End users, such as health economists using the tools, face a limited initial burden, whereas tool developers must invest significantly in building and refining AI-driven systems. For HTA evaluators, there should be no additional time burden as long as AI-generated materials adhere to existing submission formats and quality expectations. In this regard, the responsibility for ensuring accuracy and compliance remains with the submitting party, just as in the current human-driven process.

As HEOR evolves toward more dynamic and adaptive approaches, health economists will need to carefully navigate these challenges to ensure transparency and trust in the materials they develop and use.

9.2 Patient Empowerment

GenAI has the potential to democratise access to information, empowering patients to play a more active role in healthcare decision-making. Health economists could use LLMs to create personalised, easy-to-understand summaries of HEOR-based analytics, translating complex analyses into patient-friendly formats that explain the potential benefits, risks, and implications of different treatment options. Such efforts could enable more informed discussions with clinicians. By starting with smaller tasks like summarising existing analyses for lay audiences, health economists can use one of the key strengths of LLMs to support this important aspect of healthcare.

9.3 Enhanced Analytics to Support Timely Access to Medicines

The application of GenAI in HEOR has the potential to accelerate research processes, enhance analytical efficiency, and reduce human biases in evidence generation [2]. These capabilities could lead to more timely and informed decision-making in healthcare systems, including access to medicines. For example, GenAI can assist health economists by facilitating sophisticated analyses that interrogate complex datasets in new ways, thereby generating meaningful insights for clinicians and patients [70].

However, while the prospect of faster access is an aspirational goal, timeliness must be balanced with robust evaluations of safety, efficacy, and cost-effectiveness. The use of GenAI does not replace high-quality evidence generation or the critical appraisal provided by expert input In addition, challenges such as data privacy, potential model biases, and hallucinations must be carefully considered to ensure that the integration of GenAI into HEOR practices upholds the integrity and reliability of healthcare decision-making processes [73].

10 Conclusions

As with any technological disruption, GenAI is likely to introduce new opportunities and spawn new branches of health economics that are, at this moment in time, inherently unpredictable. The next few years will be pivotal in establishing best practices for GenAI, including methodological standards and ethical frameworks, that ensure GenAI supports the integrity and transparency of HEOR practices for the benefit of patients.

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